CS3249 (AY26/27 Sem 1) Assignment 1

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Project Title:
PaperLens: A Grounded Conversational Reading Companion for Non-native English Speakers to Understand English Research Papers

Project Description:
PaperLens uses goal-directed, source-grounded conversation to help learners efficiently understand and critically evaluate a research paper's motivation, methodology, evidence, contributions, and limitations.

Total word count (excluding Reference and Appendix): **1982**

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1. User Research

Before the interviews, I expected mistranslation, inaccurate summarisation, and hallucination to be the main barriers to AI-assisted paper reading. The findings revealed a broader problem:

> ***Technically oriented, non-native-English-speaking students can obtain quick summaries and isolated explanations from AI, but current interactions rarely provide a clear progression from overview to technical detail and critical evaluation, resulting in repeatedly formulated follow-up questions and fragmented answers, which is difficult for users to develop a coherent and source-grounded understanding, judge which ideas are worth drawing on, and determine how they can appropriately inform or support their own work without sacrificing efficiency.***

The participants were a Computer Science PhD student (P1), a Statistics PhD student (P2), and a Computer Science undergraduate (P3). All regularly read technical papers independently but preferred Chinese AI interactions for efficiency. They typically uploaded a paper or provided an *arXiv* link, requested an overview, and then asked focused questions for research tasks.

1.1 Key Challenges

1.1.1 Coherence of Research Logic

AI could explain individual sections clearly without showing whether the motivation, method, experiments, and conclusions formed a convincing research story. P2 particularly valued internal consistency across these elements.

1.1.2 Structure and Depth of Understanding

General summaries were often too broad for understanding algorithms, formulas, inputs and outputs, implementation details, datasets, or experimental design. P2 and P3 therefore used repeated follow-up questions to reach the required depth, but the resulting explanations could remain fragmented rather than forming a clear progression from the paper's overall purpose to its technical details and potential research use.

1.1.3 Critical Evaluation of Research Claims

P1 and P2 observed that AI often adopted the authors' framing instead of questioning novelty, assumptions, limitations, or practical value. It could also reproduce a paper's unnecessary technical complexity rather than identify its core idea. This was especially problematic in unfamiliar fields, where users lacked the background knowledge to assess plausible explanations.

1.1.4 Trade-off Between Accuracy and Efficiency

Participants did not verify every response. They returned to the source mainly for highly relevant papers, implementation details, or important figures. They recognised that AI might omit or oversimplify information, but comprehensive verification would undermine the efficiency they valued.

1.2 User Needs

Four needs should be prioritised.

1.2.1 Goal-Directed, Structured, and Layered Explanations

Learners need a clear progression from a concise overview to focused questions about methodology, limitations, and potential research use. The system should suggest relevant follow-up questions and provide explanations at increasing levels of depth, while allowing learners to choose, skip, or revisit each layer according to their immediate goal.

1.2.2 Support for Critical Evaluation

The system should help learners examine research logic, novelty, assumptions, limitations, and value rather than automatically accepting the authors' claims.

1.2.3 Explanations Connected to Original Figures

The system should explain relevant graphs or figures and their role in the paper's argument instead of replacing them with oversimplified diagrams.

1.2.4 User-controlled Conversational Support

The system should maintain context across follow-up questions. Reflection prompts should be optional, and explanations should preserve important English technical terms.

1.3 Target User Persona

Persona: Sam
Sam is a non-native-English-speaking postgraduate researcher in a technical discipline. He reads academic English but prefers concise Chinese explanations that preserve English terminology. He sets a specific goal, begins with an overview, asks focused questions, and checks the source selectively. He also needs help understanding how the paper's key elements connect, assessing its logic, novelty, assumptions, and limitations, and identifying which ideas may appropriately inform his own research. He expects a critical reading partner, not a summariser, examiner, or unquestioning assistant. (Note: Please refer to Appendix C: User Journey Map for more details.)

1.4 Proposed CUI Support

The CUI would establish the learner's reading goal, provide a brief goal-specific overview, and organise optional follow-up questions from foundational understanding to technical detail, source evidence, critical evaluation, and potential research use. Users could expand, skip, or revisit these layers while the system maintains context and links explanations to relevant passages, tables, or figures. On request, it could examine research logic, assumptions, evidence gaps, and alternative interpretations, then organise a research-use note identifying what may be relevant to the learner's work, why it is relevant, and what limitations should be considered.

The prototype therefore focuses on the goal-directed and critical reading of technical empirical papers, especially when learners must decide whether and how a paper can inform their own research.

2. CUI Design

2.1 Project Goals

2.1.1 User Learning Goals

After a focused interaction, the learner should be able to:

1. Identify which ideas, methods, evidence, or experimental approaches are relevant to their own work and justify how they may appropriately inform it;
2. Explain how a selected claim, method, or experiment relates to the paper's research question and evidence;
3. Identify at least one assumption, limitation, or unresolved uncertainty that affects whether or how the selected element can be applied.

The aim is not mastery of the entire paper, but a sufficiently accurate and critical understanding that supports informed decisions about whether and how the paper may contribute to the learner's own work.

2.1.2 CUI Design Goals

Following the human-centred principle that ***design is a function of users, tasks, and context*** (Lee, 2026b), the following goals translate the interview findings into observable interaction behaviours.

Goal 1: Keep the Interaction Relevant to the Learner's Reading and Research Goal
The CUI should establish and allow revision of the learner's goal. Guided by Grice's ***Maxim of Relation*** (Grice, 1975), responses should prioritise the current task rather than defaulting to a generic summary.

Goal 2: Support Structured and Layered Exploration
The CUI should organise exploration as a clear progression from a goal-specific overview to optional follow-up questions about key concepts, technical details, potential research use, etc. Each question should receive a concise initial answer with expandable layers for further depth. Guided by Grice's ***Maxim of Quantity*** and ***Maxim of Manner***, the system should provide sufficient information clearly without overwhelming the learner or imposing a fixed sequence.

Goal 3: Establish and Repair Common Ground
The CUI should maintain the paper, goal, selected content, and previous questions as revisable context. This reflects ***grounding***, through which participants establish mutual understanding (Clark & Brennan, 1991; Clark & Schaefer, 1989). It should confirm the current focus, clarify ambiguity, and support conversational repair without a restart (Schegloff et al., 1977).

Goal 4: Distinguish Source Evidence, Interpretation, and Uncertainty
Explanations should link to sources and distinguish the authors' statements, their evidence, the CUI's interpretation, and remaining uncertainty. Grice's ***Maxim of Quality*** discourages presenting unsupported interpretations as established facts and motivates inspectable evidence.

Goal 5: Support Critical Reading Through Mixed Initiative While Preserving Learner Control
The learner should control the interaction's direction and depth. Under ***mixed initiativ***e (Horvitz, 1999), the CUI may clarify a request or surface an assumption, evidence gap, limitation, or alternative interpretation. Such prompts remain optional, supporting rather than replacing the learner's judgment.

These goals can be evaluated by whether learners can revise their goal, control response depth, inspect evidence, distinguish source content from interpretation, repair misunderstandings, and skip prompts without disrupting the task.

2.2 Learner–CUI Roles and Responsibilities

Learner	CUI
Selects a paper and specifies a reading or research-use goal	Identifies content relevant to that goal and proposes a structured set of optional follow-up questions
Chooses a claim, method, experiment, or figure and requests detail	Provides concise, layered, source-grounded explanations

Asks follow-up questions and corrects misunderstandings	Maintains context and supports conversational repair
Determines whether and how selected elements may inform their work	Organises relevant ideas, evidence, possible applications, and limitations into a research-use note

The learner remains responsible for interpreting and evaluating the paper. The CUI provides structure, evidence, explanation, and prompts but should not present its own evaluation as objective fact.

2.3 Rationale for Using a CUI

A static summary functions mainly as discourse. This task requires dialogue because the system cannot initially know the relevant content, the learner's understanding, or the required explanatory depth, and these needs change during reading.

A CUI supports this evolving task through mixed-initiative, multi-turn interaction. The learner can set and revise the goal, while the system can clarify ambiguous requests or suggest a relevant critical question. Maintaining the paper, goal, selected content, and previous discussion supports common ground; **dialogue** state tracking provides the machine's approximation of that shared context (Lee, 2026d). If an answer addresses the wrong topic or level, conversational repair lets the learner redirect it without starting again. Grice's cooperative principles can therefore be applied dynamically according to the learner's current question.

Conversation alone, however, is insufficient for long passages, equations, tables, and figures. Following the **GUI–CUI** distinction discussed by Lee (2026b), the system should combine:

- A CUI for goal clarification, follow-up dialogue, adaptive explanation, repair, and critical prompts;
- A GUI for navigation, persistent display, figure inspection, source verification, expandable content, and note organisation.

This division allows the learner to refine what they need conversationally while keeping academic evidence visible and verifiable.

2.4 CUI Task Workflow

Please refer to Appendix D: CUI Task Workflow for more details.

3. CUI Implementation

The prototype will demonstrate one multi-turn scenario using a selected technical empirical paper, from setting a reading goal and examining grounded explanations to identifying critical issues and producing a research-use note. This prototyping approach allows the interaction to be evaluated before the complete technical system is developed (Lee, 2026c).

3.1 CUI Type and Interaction Modality

The proposed system is a **hybrid text-based CUI and graphical paper-reading interface**. It combines a paper viewer, reading-goal selector, conversation panel, expandable explanations linked to original figures, a user-initiated critical review, and a research-use note. Interaction is primarily user-initiated: the CUI may suggest a next step, but the learner chooses whether to continue, request detail, reflect, or finish.

3.2 System Components

User-interface Components

1. **Paper Viewer and Input Area**
   Accepts and displays a paper, including its figures.
2. **Reading-goal Selector**
   Records whether the user wants to evaluate relevance, understand a method, examine experiments, or review the paper critically.
3. **Conversation Panel**
   Supports Chinese or English questions while retaining the current paper context.
4. **Expandable Explanation Panel**
   Presents a concise answer with optional detail and relevant figures.
5. **Critical-review Control**
   Lets the user initiate examination of research logic, contribution, assumptions, and limitations.

Supporting System Components

1. **Document Processing and Retrieval**
   Extracts the paper and retrieves material relevant to the question.
2. **Explanation Generation**
   Produces source-grounded explanations at the requested depth and in the preferred language.
3. **Conversation Context**
   Maintains the paper, reading goal, previous questions, and current focus.
4. **Critical Analysis**
   Examines connections among motivation, methodology, experiments, and conclusions and identifies contributions, assumptions, and limitations.

3.3 Prototype Scope and Technology Stack

The prototypes will be built in **Figma** - ranging from low to high fidelity - along with a React prototype that reflects the final product but without implementing any backend logic. It will feature one manually selected paper and a scripted conversation with prepared passages, figure references, explanation layers, and critical prompts, allowing the interaction design to be tested without building the complete system.

A later implementation could be added with an API-based or open-source LLM such as Qwen, vector retrieval, structured outputs separating source content from interpretation, and a multimodal model for figures.

3.4 My Contributions and AI Involvement

Designed, implemented, and evaluated by me: interview analysis, problem definition, learning and design goals, responsibility division, interaction flow, interface states, example paper and source selection, prompt structures, critical-reading criteria, usability testing, accuracy review, and final design justification.

AI functionality represented in the system: retrieval and explanation of passages, methods, and figures; translation with preserved terminology; context maintenance; critical questions and alternative interpretations; optional feedback; and research-note organisation.

AI used during development: assistance with code drafting, debugging, prompt alternatives, test cases, microcopy, and potential edge cases.

All AI-generated code and content will be reviewed, tested, and understood before submission. Final research interpretations and design decisions will remain my responsibility.

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5. Appendix

Appendix A: User Interview Protocol

Semi-Structured Interview Protocol

Interview Aim

The **semi-structured interviews** were designed to investigate how university students use generative AI when reading English-language research papers. In particular, the interviews examined:

- How and when students incorporate generative AI into their paper-reading workflow;
- What kinds of reading and comprehension tasks they delegate to AI;
- What difficulties remain after receiving AI assistance;
- How students determine whether they have genuinely understood a paper;
- What limitations they perceive in existing AI-supported reading tools;
- What forms of conversational support may improve their understanding of academic papers.

The interviews focused on participants' actual experiences rather than evaluating their English proficiency, academic ability, or preferred learning strategies.

Interview Format and Duration

Each interview lasted approximately 20 minutes and followed five main stages.

Interview stage	Approximate duration	Main purpose
Introduction and informed consent	1 minute	Explain the study and obtain consent
Background and reading habits	3 minutes	Understand the participant's academic context and normal reading practices
Recent AI-assisted reading experience	5 minutes	Examine a specific and recent example of using AI to read a paper
Remaining difficulties after AI assistance	5 minutes	Identify unresolved comprehension problems and verification strategies
User needs and potential conversational support	4 minutes	Explore desired features and interaction preferences
Summary and closing	2 minutes	Confirm the interviewer's interpretation and collect additional comments

The interview guide contained 14 main questions. Depending on time constraints and individual participant responses, not all questions were asked in every session. Optional follow-up questions were used when a participant's initial response was brief, unclear, or raised a relevant issue requiring further exploration. As the interviews were semi-structured, the exact wording and order of follow-up questions could vary depending on each participant's experience.

Introduction and Informed Consent

At the beginning of each interview, the interviewer introduced the purpose of the study and explained that the session would last approximately 20 minutes. Participants were informed that there were no correct or incorrect answers and that the study was interested in their authentic experiences of using generative AI to read English research papers.

Participants were also informed that:

- Their names, student numbers, and other identifiable information would not be included in the report;
- Any notes or recordings would be used only to organize and analyze the research findings;
- They could decline to answer any question;
- They could stop the interview at any time.

Consent was confirmed separately for participation, audio recording, and the anonymous use of quotations. This separation allowed participants to participate in the interview without necessarily agreeing to every form of data collection or reporting.

Section 1: Academic Background and Reading Habits

The first section established the participant's academic and research context. Participants were asked about their field of study, year or level of study, reasons for reading English research papers, and the types or sections of papers they usually read.

This contextual information was important because students' reading objectives may differ substantially. For example, some students may read papers to understand a new field, while others may be searching for related work, comparing methods, preparing experiments, or identifying ideas for their own research. Their reading strategies and expectations of AI may consequently differ.

Participants were then asked when and how they normally use generative AI during paper reading. Follow-up questions explored:

- Which AI tools they use;
- Whether they upload an entire paper or provide selected passages;
- Whether they use AI for translation, summarization, concept explanation, question answering, or other purposes;
- Whether they communicate with AI in English, Chinese, or a combination of languages.

These questions were intended to identify participants' existing AI-supported workflows before discussing potential improvements.

Section 2: A Recent AI-Assisted Reading Experience

The second section focused on a specific recent experience rather than participants' general opinions about AI. Participants were asked to recall the most recent occasion on which they used generative AI to read an English research paper and explain what they needed to understand.

Where necessary, the interviewer asked whether the activity was related to coursework, class preparation, academic writing, experimentation, or independent research. Participants were also asked which parts of the paper they were reading and why they decided to use AI.

They were then invited to describe the reading process from beginning to end. Follow-up questions examined:

- What they did first;
- What information or instructions they gave the AI;
- What type of response the AI produced;
- What they did after receiving the response;
- Whether they asked follow-up questions;
- What kinds of follow-up questions they asked;
- Whether and when they returned to the original paper.

This process-oriented questioning was intended to reveal how AI was embedded in the participant's actual reading behaviour. It also helped identify transitions between reading the original paper, consulting AI, asking further questions, and applying the extracted information.

Participants were subsequently asked what they found most helpful about AI during this process. The question considered both efficiency and comprehension benefits. For example, AI might help a participant save time, identify important information, understand difficult concepts, or obtain explanations in a more accessible form.

Including this question ensured that the study did not focus exclusively on the limitations of generative AI. Understanding which functions participants already valued was also important for identifying features that should be retained in a future prototype.

Section 3: Difficulties Remaining After AI Assistance

The third section investigated comprehension difficulties that remained after participants had used AI. Participants were asked whether any content was still uncertain or not fully understood.

If a participant initially reported no remaining difficulties, neutral follow-up questions were used. For example, the interviewer asked whether any section required additional questions, whether individual sentences were

understandable but difficult to connect to the paper's overall argument, or which part would be most difficult to explain to another person.

Participants were also asked whether they had encountered cases in which an AI-generated explanation did not fully match the original paper. The wording of this question was intentionally neutral and did not assume that AI regularly produced incorrect translations or explanations. Follow-up questions explored:

- How the participant detected the inconsistency;
- Whether the AI omitted information, oversimplified the content, mistranslated a passage, or changed a logical relationship;
- How the inconsistency affected the participant's understanding;
- Whether and how the participant resolved the problem.

If participants had not encountered a clear discrepancy, they were asked whether they had received answers that appeared reasonable but whose accuracy remained uncertain. This distinction was important because a plausible but unverified explanation may influence comprehension even when the user cannot identify a specific factual error.

The interview then examined how participants judged whether they had genuinely understood a paper. Potential indicators included:

- Returning to the original text;
- Checking the methodology, results, or limitations;
- Explaining the paper in their own words;
- Evaluating whether the paper's argument was logically coherent;
- Assessing whether the AI's explanation was trustworthy;
- Discovering later that their initial understanding had been incomplete or inaccurate.

Finally, participants were asked to identify the most difficult aspect of reading research papers even with AI assistance. Depending on their responses, the interviewer could explore difficulties related to core concepts, methodology, data and results, causal reasoning, argument structure, limitations, connections between sections, terminology, or translation.

Together, these questions examined not only whether participants could understand isolated content but also whether they could integrate information across the paper and critically evaluate its argument.

Section 4: User Needs and Potential Conversational Support

The fourth section explored what participants expected from a more effective AI paper-reading tool. Participants were first invited to describe their desired support freely. Follow-up questions then examined whether the system should:

- Explain complex content in a particular manner;
- Identify the section or passage supporting an explanation;
- Distinguish between information explicitly stated by the authors and interpretations generated by the AI;
- Connect the research question, methodology, findings, and limitations;
- Remember earlier questions and maintain continuity throughout the reading session.

Participants were also asked whether they preferred receiving a complete explanation immediately or being guided through the paper through questions and shorter exchanges. Follow-up questions considered when participants wanted a rapid answer, when they were willing to engage in a longer interaction, how much interaction would become burdensome, and how the system might recognize a misunderstanding.

This part of the interview was intended to identify an appropriate balance between efficiency and active engagement. A highly interactive system might encourage deeper understanding, but it could also interrupt users who only wanted to retrieve information quickly.

A specific comprehension-checking concept was then introduced. Participants were asked to imagine that, after providing an explanation, the system invited them to restate the paper's conclusion in their own words. The system would then identify missing or inaccurate elements and provide additional clarification.

Participants were asked whether this interaction would be useful, under what circumstances it might help, and when it might become irritating. The interviewer also explored whether participants would prefer multiple-choice questions, short answers, or open-ended discussion; how often comprehension checks should occur; and whether users should be able to skip them.

Positive reactions to this proposed feature were not automatically treated as evidence of an underlying user need. Greater attention was given to participants' explanations of why the feature would or would not be useful. This helped distinguish polite approval of a hypothetical concept from evidence grounded in actual reading behaviour.

Section 5: Prioritization and Closing

Near the end of the interview, participants were asked to identify the single aspect of the current AI-assisted paper-reading experience that they would most like to improve. This prioritization question was intended to identify which problem should receive the greatest attention in the prototype.

Participants were then invited to describe any relevant experiences or difficulties that had not been covered by the earlier questions. This open-ended question reduced the risk that the interview structure would exclude issues that were important to participants.

Finally, the interviewer summarized the participant's responses, including:

- How the participant normally used AI;
- What they found most helpful;
- What difficulties remained;
- What type of support they most needed.

The participant was asked whether the summary was accurate and was encouraged to correct any misunderstandings. This final member-checking step helped improve the accuracy of the interview record and ensured that the findings reflected the participant's intended meaning.

Overall Rationale

The interview protocol progressed from broad contextual questions to a concrete recent experience, then to unresolved difficulties and potential design requirements. This structure reduced reliance on abstract opinions by asking participants to describe what they had actually done before evaluating hypothetical features.

The protocol also distinguished among three related but different outcomes of AI-supported reading:

1. Efficiency: whether AI helps users locate and process information more quickly;
2. Comprehension: whether users can explain and connect the paper's central ideas;
3. Critical understanding: whether users can evaluate the paper's reasoning, evidence, limitations, and contribution.

This distinction was important because a participant may be able to obtain a rapid summary without fully understanding or critically assessing the paper. By examining workflow, remaining uncertainty, verification practices, and desired conversational support, the interviews were designed to identify opportunities for an AI reading assistant that supports both efficient information retrieval and deeper academic understanding.

Appendix B: User Interview Transcripts

Interviewee 1: P1

Background

- Major: Computer Science
- Year of Study: 3rd-year PhD student
- Research Area: Artificial Intelligence, Computer Vision, and related fields
- Main Reading Purpose: Following research trends and identifying ideas that may inspire his own research
- Main Sections Read: Motivation and methodology

Question 1: Why do you need to read English research papers?

P1: I read papers to understand current research trends and find ideas that may inspire my own work. I pay particular attention to the motivation and methodology because these sections help me judge whether the research problem is meaningful and whether the task itself is reasonable.

Question 2: When and how do you use generative AI to help you read papers?

P1: I use generative AI in almost all of my daily paper-reading activities. I usually upload the entire paper rather than copy individual paragraphs. I use tools such as Codex, and I do not care very much about the specific model (e.g. Luna, Sol, etc.)

I normally ask AI to summarize the paper first and then ask follow-up questions, particularly about the paper's motivation. I also use it for translation, concept explanation, and general question answering. I used to ask questions in English in 2023 and 2024, but I now mostly use Chinese because current models are capable of understanding and explaining the material well in Chinese.

Question 3: Can you describe a recent experience of using AI to read papers?

P1: I use AI for paper reading almost every day. I may process approximately six papers per day. I usually ask AI to summarize them and organize the important information into a table so that I can compare the papers and identify ideas that may be useful for my own research.

Question 4: What does your typical AI-assisted reading process look like?

P1: I first ask AI to summarize the paper and then read the summary. After that, I ask questions about anything that the initial summary does not explain clearly. I mainly focus on the motivation and methodology, but I may also ask about technical details, experimental results, and whether the ablation studies are reasonable.

I rarely return to the original paper. However, I am more likely to read the original text when the paper appears especially important - for example, if it is influential, published at a top conference, open-source, or produced by a major company.

Question 5: What is the most helpful aspect of using AI?

P1: AI allows me to quickly obtain an overall understanding of a paper. It can explain things patiently, and I can discuss the paper with it through follow-up questions. It also helps me assess whether the paper's overall research story and argument are logically convincing.

Question 6: What difficulties remain after using AI?

P1: In most cases, some uncertainty remains even after using AI. When the AI cannot explain something clearly or its explanation is not logically consistent, I first try to infer the meaning myself. If I still cannot resolve the issue, I ask my classmates.

However, I can usually understand the paper's broad direction well enough to explain it to another person. The unresolved problems are more likely to concern specific reasoning or details.

Question 7: Have you encountered explanations that did not match the original paper?
P1: Yes. AI may omit information, oversimplify the content, or mistranslate parts of a paper. It is also common for an answer to sound reasonable even when I am uncertain whether it is accurate.

In practice, these problems usually do not have a major impact on me because I mainly want to understand the paper's broad direction. I am often aware that the answer may not be completely reliable, but I do not always spend time verifying it.

Question 8: What is still the most difficult part of reading papers with AI?
P1: The most difficult situation is reading a paper from an unfamiliar field because I do not know the broader research context. When I am familiar with the field, I can usually judge whether the paper and its claims are valid. Without that background knowledge, it is much harder to evaluate the paper critically.

Question 9: What support would you want from a better AI paper-reading tool?
P1: I would like the tool to compare the paper with previous work and explain how it differs from earlier papers. It should help me judge whether the contribution is sufficiently novel.

I would also like stronger critical analysis. The tool should identify limitations, questionable assumptions, and potentially unreasonable parts of the paper. It should also help me consider whether any ideas could be adapted for my own research.

Current AI tools tend to interpret papers positively and accept the authors' framing. They are not very good at identifying fundamental flaws. Their suggestions are often too general and insufficiently personalized so that it is kind of hard for me to actually learn and use things mentioned in other papers. Answers can also be unnecessarily long, overly technical, and filled with formulas or inconsistent notation. Authors sometimes make their work appear more complicated than necessary, and AI may repeat that complexity instead of simplifying it.

Question 10: Would you prefer a complete explanation or an interactive process?
P1: I do not want the tool to ask too many questions because I have limited time. Around five or six questions may be acceptable, but more than that would become annoying. A limited guided interaction seems acceptable as long as it remains efficient and does not interrupt the reading process too much.

<u>Interviewee 2: P2</u>
Background

- Major: Statistics
- Year of Study: 3rd-year PhD student
- Main Reading Purpose: Referring to and building on other researchers' work
- Main Sections Read: A quick review of the introduction, followed by closer attention to the methodology

Question 1: Why do you need to read English research papers?
P2: I mainly read papers because I need to refer to other researchers' work. I usually look through the introduction quickly and then pay more attention to the methodology.

Question 2: When and how do you use generative AI to help you read papers?
P2: I use generative AI for almost all paper-reading tasks. I normally upload the entire paper to ChatGPT and use it for translation, summarization, and concept explanation. Summarization is particularly useful for the introduction. I usually ask questions in Chinese and prefer the main explanation to be in Chinese, while keeping important technical terms in English.

Question 3: Can you describe a recent experience of using AI to read papers?

P2: I use different levels of AI assistance depending on how relevant the paper is. When initially comparing papers, I may only compare their abstracts, and using AI at this stage is optional.

If a paper addresses a problem similar to mine, I use AI to examine its introduction and methodology more closely. I want to understand what the paper improves, how it performs, what results it produces, and what advantages it has.

Question 4: What does your typical AI-assisted reading process look like?
P2: I create a research folder containing the paper, and a Markdown note that records the project process. I then add the PDF version of the paper and ask the AI whether the paper's research purpose aligns with my project.

I only read the paper myself in two cases. During the experimental phase, I read the paper to understand methods and algorithms that may be relevant to implementation. During the writing phase, I return to papers when preparing the related-work section.

Question 5: What is the most helpful aspect of using AI?
P2: AI makes my overall research process much faster. It may make paper-related work approximately five times faster, and coding tasks can sometimes become dramatically faster.

AI also helps me manage the research project, including documents, folders, notes, and progress. In addition, I use it to help write essays and research papers.

Question 6: What difficulties remain after using AI?
P2: Almost every paper requires follow-up questions. I need to ask about individual components, especially the code, algorithms, and technical implementation. A general summary is not enough when I need to understand how the method actually works.

Question 7: Have you encountered explanations that did not match the original paper?
P2: Yes. Around 2025, I encountered a long-context problem in which the AI mixed content from Paper A into its explanation of Paper B. I noticed the error because I remembered what the papers contained.

When this happens, I read the relevant parts myself or begin a new conversation to prevent information from different papers from being mixed together.

For paper reading, this kind of problem is relatively manageable. In coding tasks, however, the context must be defined very clearly—for example, whether the code is local or remote and what scale should be used for a smoke test. I have to clarify the requirements before AI starts doing things, otherwise it could be very troublesome.

Question 7: How do you decide whether you have truly understood a paper?
P2: I consider myself to understand the paper when its methodology, experiments, results, and overall logic are internally consistent.

There have been situations in which I initially thought I understood a paper but later realized that I had missed important details, such as the code structure or the exact inputs and outputs. When that happens, I continue asking the AI questions.

Question 8: What is still the most difficult part of reading papers with AI?
P2: The most difficult part is understanding the rationale behind the paper's overall story. The introduction and methodology should form a consistent argument, but sometimes they do not connect clearly.

This may be a problem with how the paper is written rather than a limitation of the AI. Even if the AI explains each section, the underlying research story may still be unclear or unconvincing.

Question 9: What support would you want from a better AI paper-reading tool?
P2: Current AI tools are already very good at translation and explanation. The larger problem is that they tend to accept the paper's framing and cater too much to the authors' perspective.

AI may be able to criticize individual methodological choices, but it is weaker at high-level critical judgment. It does not always have strong "paper taste"- the ability to judge whether a research question, contribution, or overall story is genuinely valuable.

Question 10: Would you prefer a complete explanation or an interactive process?
P2: I prefer an ask-and-explain process rather than receiving a complete explanation all at once. A complete explanation is usually too long. It is more efficient for me to ask about specific issues and receive focused answers.

Question 11: Would it help if the system asked you to explain the paper in your own words?
P2: I would prefer the user to ask questions rather than have the AI test the user. I do not necessarily want the system to initiate comprehension checks or ask me to restate the conclusion.

Question 12: If you could improve one aspect of the current experience, what would it be?
P2: I would improve the AI's critical thinking and paper taste. It should be better at judging the value, novelty, and overall quality of a paper rather than simply explaining or supporting the authors' argument.

Interviewee 3: P3
Background

- Major: Computer Science
- Year of Study: 3rd-year undergraduate student
- Research Area: AI Memory- and audio-related research
- Main Reading Purpose: Finding and writing about related work for their own research
- Main Sections Read: Abstract and introduction first, followed by the experiments when necessary

Question 1: Why do you need to read English research papers?
P3: I read English papers mainly because I need to write the related-work section of my own paper. My research concerns memory- and audio-related topics.

I usually begin with the abstract and introduction. I look at the experimental section later if the paper appears relevant.

Question 2: When and how do you use generative AI to help you read papers?
P3: I do not use AI when I only need to look up a baseline value. I use it when I need to understand important work in a field or learn the background of an unfamiliar topic.

I normally provide Codex with an arXiv link rather than manually uploading or copying parts of the paper. I ask it to identify the paper's main innovation, explain its position within the field, examine its citations, and help me quickly judge whether the paper is suitable for inclusion in my related-work section.

I ask questions in Chinese and prefer answers in Chinese.

Question 3: Can you describe a recent experience of using AI to read papers?

P3: Recently, I used AI while preparing the experimental section of my own paper. I wanted to refer to how other papers organized their experiments. I looked at their main and secondary tasks, graphs, number of figures, and the way they presented the overall research story.

Question 4: What does your typical AI-assisted reading process look like?
P3: I first give the AI a link and ask what the paper is mainly about. I then ask it to explain the experimental structure in the simplest possible way.

The first answer is often too general, so I ask follow-up questions about datasets, baselines, and other specific elements. If the answer is too verbose, I ask the AI to condense it and present the information as a numbered list.

After understanding the experimental structure, I return to my own work and ask the AI to help me design a similar experimental framework. I normally define the overall framework first and then develop the individual experiments.

I also return to the original paper while using AI, particularly when I need to examine non-textual information such as graphs and figures.

Question 5: What is the most helpful aspect of using AI?
P3: The main benefit is efficiency. I can read English, but reading every sentence and technical detail takes too much time. AI allows me to focus on the paper's core information without reading every part in full.

Question 6: What difficulties remain after using AI?
P3: I usually do not aim to understand the entire paper. For my immediate purpose, understanding the experimental section may be sufficient. It is unrealistic to fully understand a paper within a very short period.

The methodology remains difficult, especially when it contains complex formulas. AI can summarize the method, but I may still not fully understand how it works.

Question 7: Have you encountered explanations that did not match the original paper?
P3: I have not encountered major cases in which the AI's explanation directly contradicted the paper. It may omit information, but its translation is generally accurate.

The larger problem is that answers are often unnecessarily verbose and frustrating to read. I usually need to state clearly in the prompt that I want a concise response.

Because I mainly use AI for summarization and identifying the paper's direction, hallucination has not been a significant problem for me.

Question 8: How do you decide whether you have truly understood a paper?
P3: I generally read through the AI's answer rather than explain the paper again in my own words. Since my goal is often to obtain specific information rather than master the entire paper, I do not always perform an explicit comprehension check.

Question 9: What is still the most difficult part of reading papers with AI?
P3: The most difficult part is understanding methodology that involves complex formulas. AI explanations are more effective for applied or industry papers. They are less effective for fundamental technical papers, such as work on optimizing the Transformer architecture.

However, I think this difficulty mainly comes from the complexity of the paper itself rather than from the AI.

Question 10: What support would you want from a better AI paper-reading tool?
P3: Current tools such as Codex already work very well for text-based explanations. I have tried tools that convert an introduction into a diagram, but the diagrams were too simple and condensed the information too much. They were not more useful than a clear plain-text explanation.

A useful additional feature would be to use the paper's original graphs and figures and provide explanations alongside them. This would be more valuable than generating a new but oversimplified diagram.

Question 11: Would you prefer a complete explanation or an interactive process?
P3: User should take the initiative to ask questions to AI, otherwise AI won't know what to answer!

Question 12: Would it help if the system asked you to explain the paper in your own words?
P3: Yes, I think that would be helpful, provided the system doesn't sound like a strict examiner. It should adopt a supportive, exploratory tone - as if the user and AI are exploring the paper together.

Current AI tools often tend to over-agree or echo the user's opinions, which breaks the immersion. I once had an experience where I claimed 'A is correct' and the AI agreed, but as soon as I switched to 'A is wrong,' it blindly agreed with that as well. It was really frustrating. A thoughtful comprehension check would make the AI feel less like a yes-man and foster a much more genuine discussion.

Question 13: If you could improve one aspect of the current experience, what would it be?
P3: I would add explanations that are directly connected to the paper's graphs and figures. This could make complex information easier to understand while preserving more of the paper's original detail.

Appendix C: User Journey Map



Fig. 1. User Journey Map. Refer to FigJam for more details.

Appendix D: CUI Task Workflow

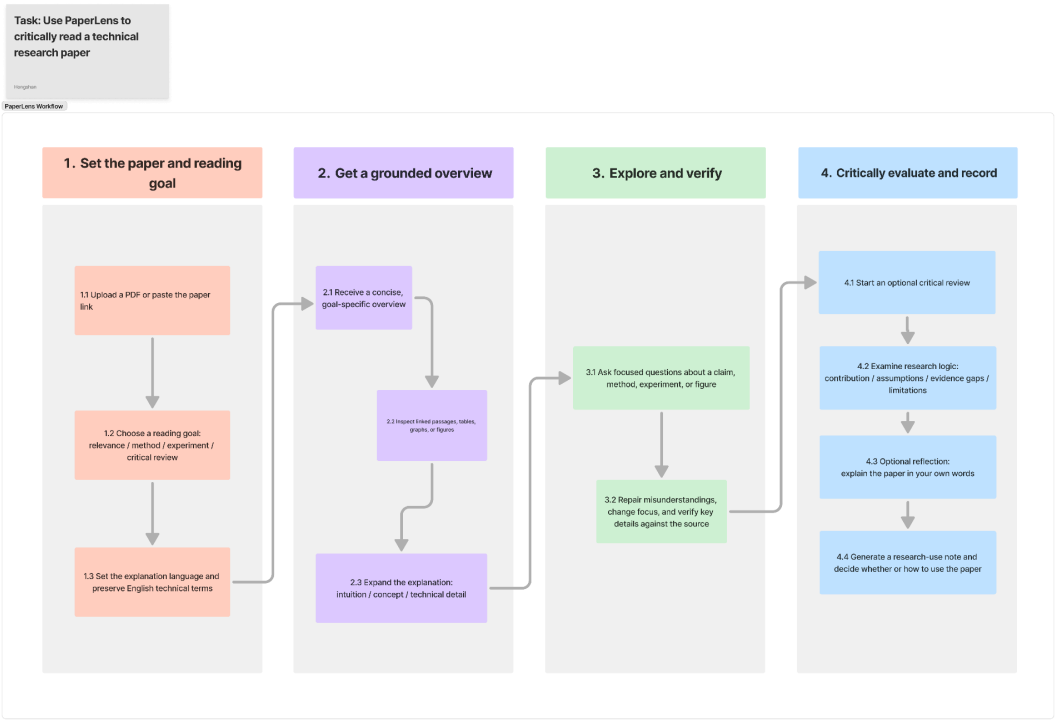


Fig. 2. ***PaperLens*** CUI task workflow diagram. Refer to FigJam for more details.

Appendix E: Prototypes

To **iteratively** refine the conversational user interface, **four development milestones** were executed: (i) low-fidelity paper wireframes, (ii) mid-fidelity static mockups in **Figma**, (iii) high-fidelity interactive designs in Figma, and (iv) a **React-based** web prototype (**without** backend technical implementation).

The visual evolution and live implementation across these fidelity stages are summarized below (see **Figure 3–6** for interface screenshots and links):

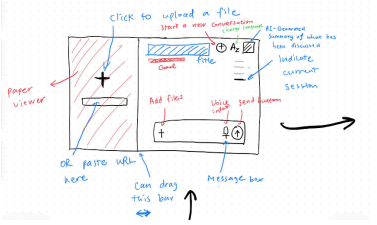


Fig. 1. Main page

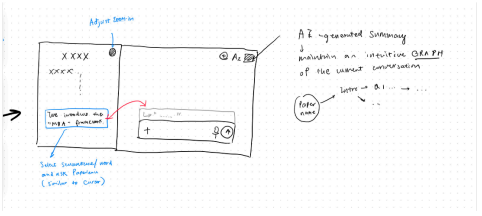


Fig. 2. Select a sentence and ask a question

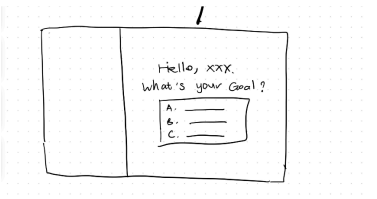


Fig. 3. Opening Page

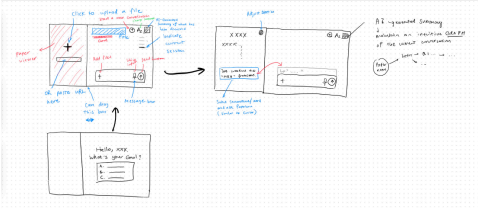


Fig. 4. Overview

Fig. 3. Low-fidelity prototype of ***PaperLens***, developed by Figma



Fig. 4. Mid-fidelity prototype of ***PaperLens***, developed by Figma

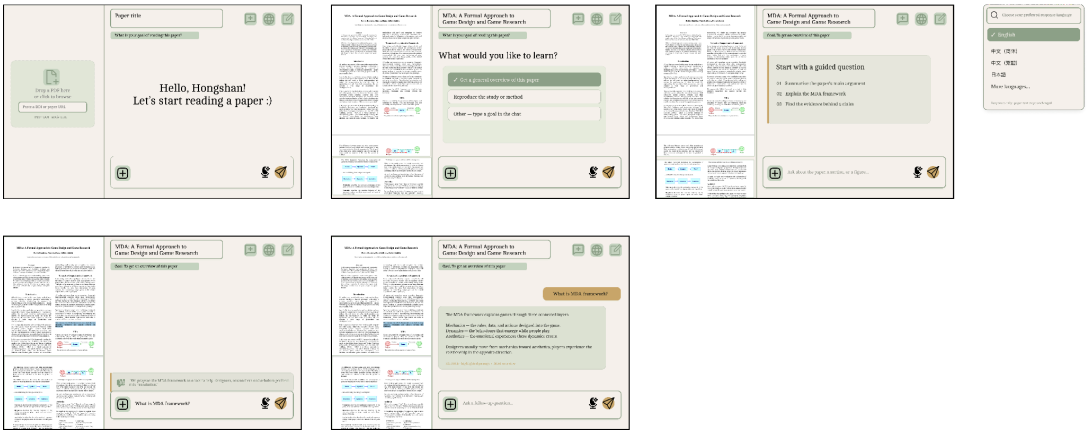


Fig. 5. High-fidelity prototype of ***PaperLens***, developed by Figma

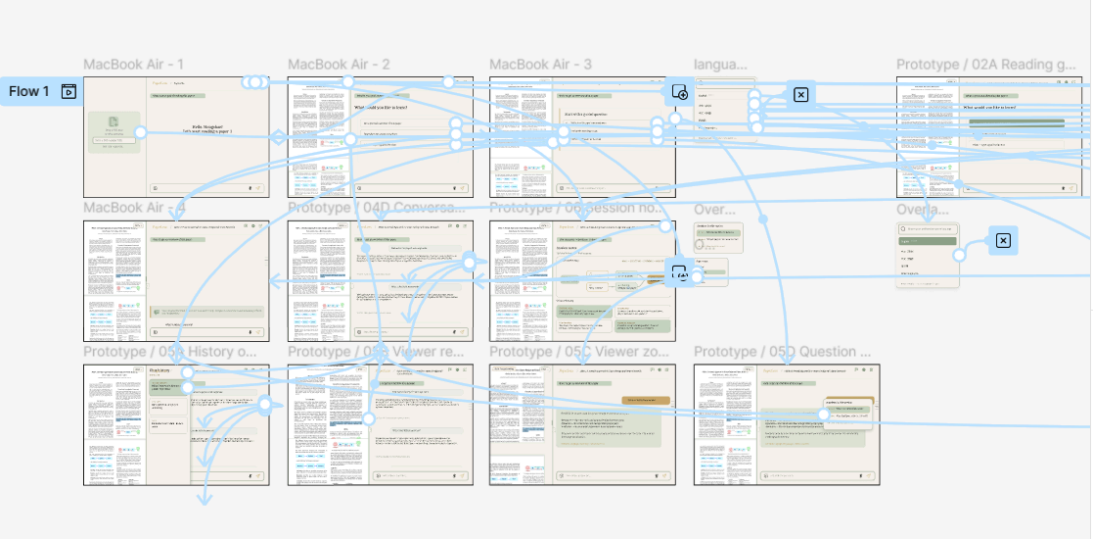


Fig. 6. High-fidelity **interactive** prototype of ***PaperLens***, developed by Figma

The **high-fidelity interactive prototype** can be viewed here.

In addition to using Figma, I implemented the final prototype as a **functional web application** (as shown below). The **source code** is available in the following GitHub repository, and the **web application** is deployed here.

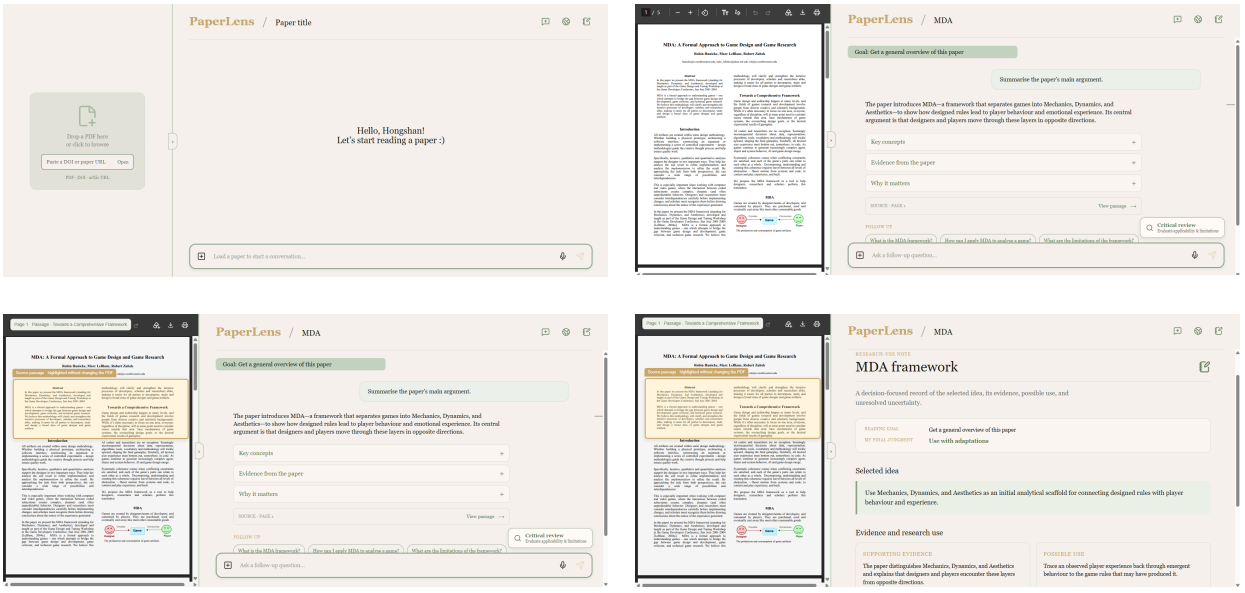


Fig. 6. Screenshots taken from the React-based web page.